HCL DX Compose – Image Mirroring from/to Harbor

Executive Summary

Enterprise OpenShift and Kubernetes platforms commonly operate with **internal container registries** to enforce security, compliance, and operational control. Although HCL provides a publicly accessible registry for **HCL Digital Experience (DX)** images, many customers—especially those running in **restricted or air-gapped environments**—must mirror these images into their own registries.

This document describes a **simple and auditable workflow** to:

- Download HCL DX Compose container images as an offline archive
- Upload them into an **internal Harbor registry**
- Enable their consumption during platform deployments

Multiple mirroring strategies are briefly discussed, with guidance on when to use each one. The guide intentionally focuses on a **low-complexity, offline approach**, suitable for a wide range of customer environments, while still allowing future evolution toward automated mirroring solutions.

Background / Context

Although **HCL provides a publicly accessible Harbor registry** that customers can use to automatically pull the container images required to deploy **HCL Digital Experience (DX)**, many enterprise environments rely on **internal container registries**.

This approach allows organizations to:

- Enforce **corporate security and compliance policies**
- Control image provenance and vulnerability scanning
- Operate in **restricted or fully air-gapped OpenShift environments**
- Reduce external network dependencies during cluster operations

In such scenarios, container images must be **mirrored from the vendor registry into a customer-managed registry**.

Several supported strategies exist to achieve this:

- **Transparent mirroring using oc-mirror / oc2mirror**, allowing a controlled and declarative mirror from external registries into internal ones.This approach is well suited for **OpenShift environments** and can be combined with **registry replication mechanisms** to ensure **long-term maintenance and lifecycle management** of container images.
- **Direct registry-to-registry copy** using tools such as **skopeo**
- **Harbor replication rules**, when the customer also uses Harbor, to automatically mirror vendor images and keep them synchronized over time
- **Offline distribution**, by downloading images as an archive (ZIP) and uploading them into the internal registry

This guide focuses on the **offline distribution approach**, demonstrating how to download HCL DX Compose images as an archive and upload them into a **Harbor registry**.

High-Level Flow

```
Parse error on line 1:
flowchart LR    A[H
^
Expecting 'NEWLINE', 'SPACE', 'GRAPH', got 'ALPHA'
```

Workflow Overview

The process consists of:

1. Downloading **HCL DX Compose** from the HCL Software Portal
2. Extracting the container images locally
3. Uploading the images to an **internal Harbor registry**
4. Consuming the images from the registry during deployment

The example used in this guide references **DX Compose version 9.5.2-232**.

Step-by-Step Instructions

1. Access the Download Portal

Navigate to the official HCL Software download page for DX Compose:

```
https://my.hcltechsw.com/downloads/dx/compose
```

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2. Select Version

Select the appropriate release version for your target environment.

In this example:

- **Major version:** 9.5
- **Build:** 9.5.2-232
- **Release date:** December 15, 2025

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3. Generate the Download Link

Locate the archive:

- **File name:** `hcl-compose-kubernetes-CF232`
- **Size:** ~6.88 GB

Generate a **temporary, pre-authenticated URL** (valid for 60 minutes) using the *Copy temporary link* option.

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4. Download the Archive

Download the archive to a bastion host or administrative node with network access.

```
wget -O hcl-cf232.zip "https://cdn.hcltechsw.com/hcl-compose-kubernetes-CF232.zip?Object=..."
```

> The URL shown above is truncated. Use the full temporary link generated in the previous step.

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5. Extract the Images

Extract the archive into a working directory:

```
unzip hcl-cf232.zip -d images
```

This directory will contain the container images and metadata required for upload.

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6. Upload Images to Internal Registry (i.e, Local Harbor)

Once the images are available locally, they can be uploaded manually to the internal registry using standard container tooling (e.g. **podman or skopeo**).

In addition, it is recommended to upload the existing Helm charts associated with the DX deployment to the internal repository (or chart registry), ensuring that both container images and deployment artifacts are available locally and aligned with the same version lifecycle.

To simplify and automate this process, an **optional helper script** (`upload_harbor.sh`) is provided.

This script iterates over the extracted images (**tar.gz files**) and performs the load, tag, and push operations automatically. It also uploads the Helm charts required for the installation (**tgz files**) to the same location. The script takes 4 parameters: the URL of your internal registry, the authentication username, the password, and the registry project (or directory). Sample Script Example usage:

```
$ ./upload_harbor.sh <MY_REGISTRY> <USER> <PASSWORD> <PROJECT>
```

> **Note:** If the registry requires authentication to download images, you will also need to create a secret (as described later in the installation guide).

Access and Prerequisites Note

Depending on the image distribution method selected, **different access requirements apply**:

- If you plan to use **online or registry-based methods** (such as oc-mirror, skopeo copy, or Harbor replication), the customer must have:
  - **Network access** to the HCL container registry
  - Valid **credentials for the HCL Harbor registry** located at:
    - https://hclcr.io/

- If you plan to use the **offline ZIP distribution method** described in this guide, the customer must have:
  - Access to the **MyHCL Software Portal**
  - Valid **MyHCL credentials** to download the DX Compose archive from:
    - https://my.hcltechsw.com/

The **official and detailed HCL procedure** for downloading and handling DX container images is documented at:

- https://help.hcl-software.com/digital-experience/9.5/CF232/get_started/download/harbor_container_registry/

This guide complements the official documentation by providing a **simplified, operational example** focused on customer-managed registries.

Decision Guide: oc-mirror vs Offline ZIP

When choosing how to mirror HCL DX Compose images into an internal registry, the decision mainly depends on **environment connectivity, operational maturity, and long-term maintenance needs**.

Use **oc-mirror / oc2mirror** when:

- You are running **OpenShift** and want a **declarative, supported mirroring workflow**
- The environment has **controlled but available access** to external registries (or via a bastion)
- You want **transparent image resolution** without changing deployment manifests
- **Long-term maintenance** is required, including:
  - Periodic updates
  - Consistent image lifecycle management
  - Integration with **registry replication** strategies
- The platform team is comfortable operating OpenShift-native tooling

Recommended for: Production platforms, regulated environments with structured operations, and clusters that require continuous updates.

Use **Offline ZIP distribution** when:

- The environment is **fully air-gapped** or has **no outbound connectivity**
- Image transfer must be performed via **manual or controlled file exchange**
- The goal is a **one-time or infrequent deployment**
- Operational simplicity is preferred over automation
- The target audience includes **non-platform specialists**

Recommended for: Proof-of-concepts, isolated environments, initial bootstrap phases, or customer teams with limited OpenShift experience.

In this guide, the **Offline ZIP approach** is used as it provides a **simple, explicit, and easy-to-audit workflow**, while still allowing a future transition to oc-mirror or registry replication if needed.

Security Considerations

- Treat robot tokens as **secrets**
- Avoid hardcoding credentials in scripts or repositories
- Prefer environment variables or secret management solutions
- Rotate credentials if exposure is suspected
